

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: USB3 Gen T Link Ready Handshake
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Corrects when a Gen T Port enters U0 during the course of a Link Ready Handshake.

Benefits as a result of the changes:

Reduces the ambiguity that currently exists if one of the port partners sends a packet after it thinks the link is in U0 and the other is still waiting to complete the Link Ready Handshake.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

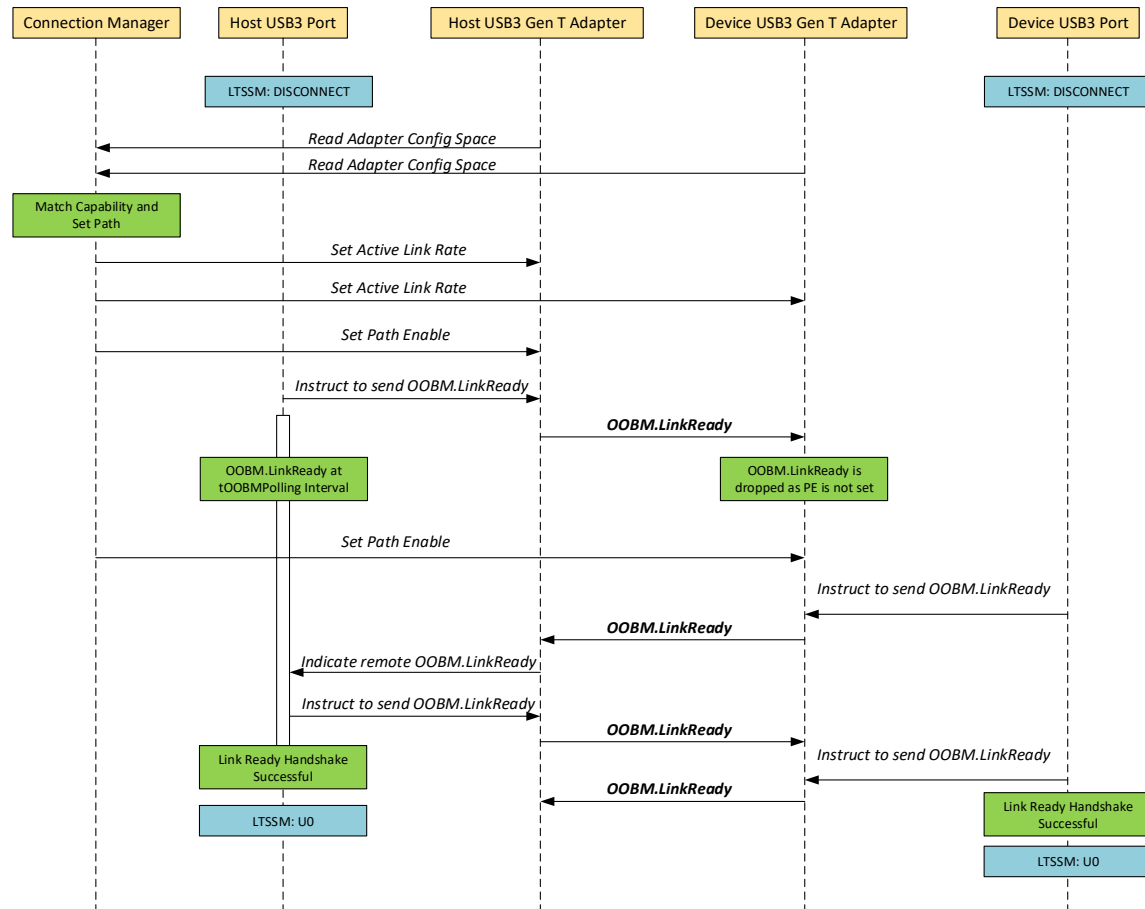
An analysis of the compliance testing implications:
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None

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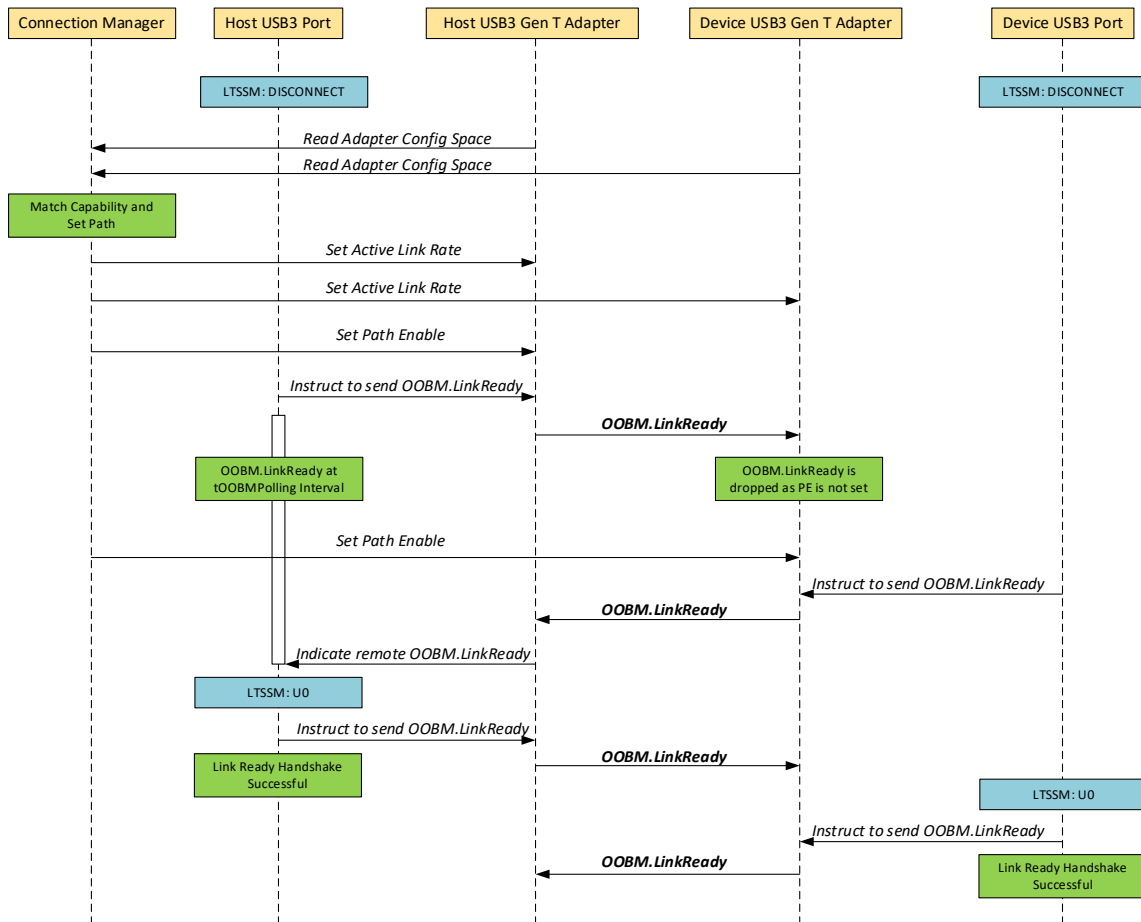
Actual Change

(a). Figure 9 21. USB3 Gen T Link Ready Handshake Sequence From:



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To:



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(b). Section 9.4.1.6.2. LTSSM.U0

From:

A USB3 Gen T Port initiates a Link Ready Handshake by sending an OOBM.LinkReady message to the link partner. The USB3 Gen T Port shall continue to send OOBM.LinkReady messages at *tOOBMPolling* intervals until it receives an OOBM.LinkReady from the link partner. After receiving at least one OOBM.LinkReady from the link partner, a USB3 Gen T Port shall send one last OOBM.LinkReady. The last OOBM.LinkReady shall be sent no later than *tOOBMPolling* Max time after receiving an OOBM.LinkReady from the link partner.

To:

A USB3 Gen T Port initiates a Link Ready Handshake by sending an OOBM.LinkReady message to the link partner. The USB3 Gen T Port shall continue to send OOBM.LinkReady messages at *tOOBMPolling* intervals until it receives an OOBM.LinkReady from the link partner. After receiving at least one OOBM.LinkReady from the link partner, a USB3 Gen T Port shall ~~enter LTSSM.U0~~ send one last OOBM.LinkReady. The USB3 Gen T Port shall send one last OOBM.LinkReady as the first packet after entry into LTSSM.U0. The last OOBM.LinkReady shall be sent no later than *tOOBMPolling* Max time after ~~entering LTSSM.U0~~ receiving an OOBM.LinkReady from the link partner.

Note: A USB3 Gen T Port shall ignore all OOBM.LinkReady messages when it is in LTSSM.U0.

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(c). Section 9.4.1.6.3. LTSSM.U3

From:

Exit from LTSSM.U3 may be initiated by a Remote Wake capable device or by the Host. A Remote Wake capable Internal USB3 Gen T Component shall request a wake by sending an OOBM.U3Resume message while staying in U3. It shall send OOBM.U3Resume at tOOBMPolling intervals until it receives an OOBM.LinkReady from the Host. Once it receives the OOBM.LinkReady, it shall complete the Link Ready handshake and transition into LTSSM.U0.

Note: There is no timing requirement on when a Host sends the OOBM.LinkReady as it is dependent on software.

A Host-initiated LTSSM.U3 exit is directed by system software. When directed to do so by system software, a USB3 Gen T downstream port shall initiate the LTSSM.U3 exit flow by sending OOBM.LinkReady to the link partner at tOOBMPolling intervals. It shall then wait for an OOBM.LinkReady from the link partner and complete the Link Ready handshake and transition into LTSSM.U0.

To:

Exit from LTSSM.U3 may be initiated by a Remote Wake capable device or by the Host. A Remote Wake capable Internal USB3 Gen T Component shall request a wake by sending an OOBM.U3Resume message while staying in U3. It shall send OOBM.U3Resume at tOOBMPolling intervals until it receives an OOBM.LinkReady from the Host. Once it receives the OOBM.LinkReady, it shall complete the Link Ready handshake as described in Section 9.4.1.6.2~~and transition into LTSSM.U0.~~

Note: There is no timing requirement on when a Host sends the OOBM.LinkReady as it is dependent on software.

A Host-initiated LTSSM.U3 exit is directed by system software. When directed to do so by system software, a USB3 Gen T downstream port shall initiate the LTSSM.U3 exit by initiating a Link Ready handshake as described in Section 9.4.1.6.2~~flow by sending OOBM.LinkReady to the link partner at tOOBMPolling intervals. It shall then wait for an OOBM.LinkReady from the link partner and complete the Link Ready handshake and transition into LTSSM.U0.~~

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(d). Section 9.4.1.6.4. LTSSM.U2

From:

If the Internal USB3 Gen T Component has data to transfer, it shall wake up the USB3 Gen T Component that it is connected to by sending *OOBM.LinkReady* to its link partner at *tOOBMPolling* intervals. It shall then wait for an *OOBM.LinkReady* from the link partner and complete the Link Ready handshake and transition into LTSSM.U0.

To:

If the Internal USB3 Gen T Component has data to transfer, it shall wake up the USB3 Gen T Component that it is connected to by initiating a Link Ready handshake with its link partner as described in Section 9.4.1.6.2~~sending *OOBM.LinkReady* to its link partner at *tOOBMPolling* intervals.~~ ~~It shall then wait for an *OOBM.LinkReady* from the link partner and complete the Link Ready handshake and transition into LTSSM.U0.~~